Chi-Wei Lee

NeuroAI — predictive coding, associative memory, and Bayesian inference in brains and machines
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Education

National Tsing Hua University – Taiwan 2022 – 2026

B.S. in Physics and Electrical Engineering & Computer Science (Double Major, AI Track)

- **Mei Yi-Chi Memorial Medal** (2026) — NTHU's highest honor for graduating students: 1 of 8 university-wide, sole recipient in the College of Science, and the only Physics recipient in nine years.
- GPA 4.10/4.30, departmental rank 1 of 72 (Spring 2024 semester). Teaching Assistant, *Introduction to Machine Learning* (Fall 2025).

Publications & Manuscripts

Predictive Coding with a Sequential Memory-Driven Bayesian Denoiser Under review

C.-W. Lee (co-first author) et al. A Tweedie–Hopfield correspondence identifies associative memory as the Bayesian denoiser predictive coding implicitly requires, yielding HOPE — one energy unifying hierarchical inference, temporal prediction, and layer-wise memory. Beats predictive-coding and Hopfield baselines on Bouncing Moving-MNIST and UCF-101 recall using only local Hebbian updates (no BPTT or weight transport).

Langevin Temporal Predictive Coding Lifts Modes to Multimodal Densities Under review

C.-W. Lee (co-first author) et al. Replaces the deterministic MAP inference of temporal predictive coding with preconditioned Langevin dynamics, making the filtering posterior the flow's unique stationary density. Lifts mode coverage from 19.4% to 71.3% (3.67×) and cuts predictive MAE 16.7% (22.3% on naturally multimodal tasks) over twelve UCI benchmarks.

MatrixQR: Matrix-Guided Refinement for Decoupling Reliability Control from Creative Edits in QR Code Generation 2025

C.-W. Lee, Y.-H. Hsueh, J.-A. Guo, J.-W. Liao, P.-Y. Chiu, J.-C. Chen. *TAAI 2025* — accepted poster (extended abstract).

Classifying Vibration Modes Generated by the Michelson Interferometer Using Machine Learning Methods 2024

X.-H. Tsai, A. A.-C. Yeh, C.-H. Lu, S.-Y. Chou, S.-W. Wang, **C.-W. Lee**, P.-H. Lee. *Journal of Modern Physics* 15(12). doi:10.4236/jmp.2024.1512087.

Research Experience

Human-Centered Machine Intelligence Lab, National Tsing Hua University – Taiwan 2026 – present

Undergraduate Researcher · PI: Prof. Po-Chih Kuo

- Reconstructing 3D spatial representations from fMRI signals with generative decoding models; manuscript in preparation.

University of California, Los Angeles · Taiwan Global Pathfinders Initiative – Los Angeles, CA Jul – Sep 2026

Visiting Research Delegate, AI × Brain and Neuroscience Track

- One of 9 delegates selected nationally for a 60-day research placement at UCLA, funded by Taiwan's Ministry of Education.

Research Center for Information Technology Innovation, Academia Sinica – Taiwan 2025 – present

Summer Intern, then Part-time Research Assistant · Advisor: Dr. Jun-Cheng Chen

- Led diffusion-based aesthetic QR-code generation, decoupling creative editing from scan-reliability control (MatrixQR).

National Tsing Hua University – Taiwan 2024 – 2025

Undergraduate Researcher, Medical Imaging

- Built attention-transfer models for chest X-ray classification that target pathology-subgroup bias, long-tail label distributions, and class imbalance.

Honors & Awards

- **2nd of 600+ teams worldwide** (Overall Track, 2025) and **3rd worldwide** (Climate Prediction Track, 2026) — U.S. NSF HDR Machine Learning Challenge. Team lead (6) on autoencoder- and diffusion-based anomaly detection; presented at the AAAI 2025 Workshop.
- **1st place** — Mei-Chu Hackathon, Maker Track (2025), one of Taiwan's largest student hackathons: a wearable-and-app system for dementia walking-safety monitoring.
- **Gold Medal** — iGEM 2023, Dry Lab Lead: ML colorectal-cancer screening plus an Arduino-integrated diagnostic device; presented in Paris.

Technical Skills

Python, PyTorch, JAX/NumPyro, CUDA · diffusion and energy-based models · MCMC/Langevin inference · fMRI decoding